



Agreed at Four Days: A Three-Round Delphi Study of Household Leftover Discard Thresholds, Tested Against a Withheld-Feedback Round and a 96-Household Fridge Diary

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Abstract. Households decide daily whether a container of leftovers is still food, and no standardised threshold governs the judgement. We convened a four-stratum expert panel ($n = 31$) and ran a three-round modified Delphi to establish one, with a consensus criterion fixed in advance. The criterion was satisfied at round three: the interquartile range of nominated thresholds contracted from seven days to one, and 26 of 31 panelists sat within a day of the panel median. The median itself moved two days across the same interval. Coding the free-text rationale accompanying each nomination indicates that panellists revised their figures far more readily than their reasons — 24 of 31 changed the number between rounds one and two while 7 changed the stated basis for it — a pattern we term revision without reconsideration. A pre-registered fourth round, administered six weeks later with the group median withheld from a randomly assigned half of the panel, found the contracted spread did not persist without the feedback that produced it. A parallel 28-day fridge diary kept by 96 households (1,140 logged containers) records a bimodal discard distribution with a median of eight days; 21% of containers were discarded within a day of the ratified threshold. The ratified figure has since been tabled and entered on the Horizon Register as a settled definition. Our results suggest the procedure converges reliably, and we recommend it be retained.

Introduction

A container of leftovers in a domestic refrigerator occupies an unusual position in household governance: everyone makes a decision about it, the decision has a determinate outcome, and no institution has ever specified when the decision should be made. Date labelling covers the sealed and the purchased [1]; the cooked, decanted, and re-stacked remainder of last night's meal falls outside every labelling regime and is left to the judgement of whoever opens the door.

That judgement has been studied as behaviour. Household food waste is consistently characterised as the product of many interacting domestic practices rather than a single decision

point [2], and sociological accounts locate it in the ordinary organisation of provisioning and eating rather than in carelessness [3]. Households have been shown to negotiate food-safety guidance against waste-reduction norms as competing anxieties [4], and both survey [5] and waste-audit [6] work has mapped the attitudes and volumes involved. What this literature does not supply, and does not set out to supply, is a threshold: a number of days after which a container may be recorded as no longer food.

We take the absence of that number to be the tractable part of the problem. Where a quantity is not directly measurable but is judged repeatedly by practitioners, structured expert elicit-

tion is the conventional instrument [7], [8], and its application to appropriateness judgements is well established [9]. We accordingly convened a panel and ran a three-round modified Delphi to fix the threshold, alongside a household diary arm against which the resulting figure could be checked.

We make three contributions, each of which we would characterise as provisional:

- We report a ratified four-day threshold for household leftover discard, obtained under a consensus criterion specified before the first round and met at round three.
- We show that the panel’s convergence toward that threshold consists substantially of numerical revision unaccompanied by any change in stated reasoning, a pattern that may be more general than this instrument.
- We compare the ratified threshold against 1,140 diary-logged discard events and observe that the two describe different distributions, though the divergence may reflect the diary period rather than the threshold.

Background

The Delphi technique elicits a group estimate through successive anonymous rounds in which each participant sees an aggregate of the previous round before revising [7]. Guidance on panel constitution, round count, and stopping rules is extensive [8], [10], and the technique has been reviewed critically as a forecasting instrument [11]. Alternative structured formats, including the nominal group technique, trade anonymity for face-to-face reconciliation [12].

The definition of consensus is the technique’s known weak seam. A systematic review of 100 published Delphi studies found that most did not define consensus explicitly, and recommended that the criterion be specified in advance and reported [13]. We follow that recommendation.

A separate literature bears directly on what controlled feedback does to a panel. Aggregated independent estimates can be strikingly accurate [14], but the accuracy depends on the estimates remaining independent: exposing participants to others’ estimates during iterated estimation narrows the diversity of opinion while raising confidence [15], and an artificially seeded initial signal has been shown to produce persistent herding in subsequent ratings [16]. A procedure that both feeds back a median and treats the

resulting contraction as evidence of agreement is therefore measuring something whose provenance is worth establishing.

Prior work at this institution is relevant on the second point. A coding study of Horizon Register scan entries and their matched committee minutes found hedges and open questions falling to a median of zero in translation, with 71% of minutes retaining none at all [17]. On the domestic side, an unannounced audit of shared-kitchen clear-outs found reported labelling compliance rising from 61% to 83% with no reliable association to the weight of food actually discarded [18] — a precedent for the gap between a governed figure and a bin.

Method

Panel. Thirty-four panellists were invited across four strata; 31 completed all three rounds (Table 1). Strata were defined by the basis of the panellist’s expertise rather than by employer, and no panellist was informed of the stratum boundaries.

Stratum	Invited	Completed
Food handling	10	9
Household practice	11	10
Procurement	7	6
Institutional	6	6
Total	34	31

Table 1: Panel composition and retention across the three rounds.

Rounds. In round one, each panellist nominated a single integer threshold — the number of days a cooked dish may remain in refrigerated storage before it should be recorded as no longer food — and supplied a free-text rationale. Rounds two and three returned to each panellist the anonymised group median and interquartile range alongside their own prior nomination, and invited re-nomination with a fresh rationale. Rounds were separated by eleven days.

Consensus criterion. Following [13], the criterion was fixed before round one and lodged with the convening secretariat. Consensus was defined as both conditions holding simultaneously:

$$\text{IQR} \leq 1 \text{ day} \quad \wedge \quad \frac{|\{p : |x_p - \tilde{x}| \leq 1\}|}{n} \geq 0.70$$

where x_p is panellist p 's nomination and $\bar{x}$ the panel median. No maximum round count was set.

Rationale coding. Every free-text rationale was independently coded by two raters against a six-category scheme (sensory, elapsed-time heuristic, dish-composition, storage-condition, deference-to-guidance, deference-to-panel), blind to round and to the nominated figure. Agreement was substantial (Cohen's kappa = 0.79); disagreements were resolved by discussion. A panellist was scored as having changed their reasoning between rounds if their primary code changed.

Withheld-feedback round. A fourth round was pre-registered and administered six weeks after ratification. Panellists were randomly assigned 1:1 to receive the round-three group median as usual (fed arm, $n = 16$) or to re-nominate with no aggregate shown (withheld arm, $n = 15$).

Diary arm. Ninety-six households, recruited independently of the panel, kept a 28-day fridge diary, photographing and dating each container of cooked leftovers at storage and again at discard or consumption. Containers consumed rather than discarded were logged and excluded from the discard distribution, leaving 1,140 discard events.

Results

The criterion was met at round three. The distribution of nominations contracted sharply and monotonically (Figure 1). The interquartile range fell from seven days at round one to three at round two and one at round three; 26 of 31 panellists (84%) sat within a day of the round-three median of four days, satisfying both conditions. The panel median moved from six days to five to four across the same interval. Contraction therefore exceeded relocation by roughly three to one.

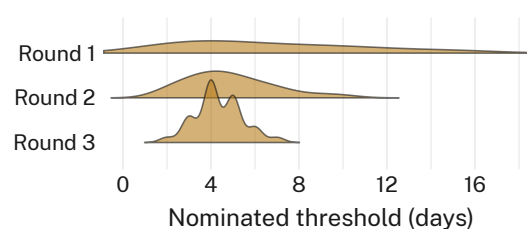


Figure 1: Nominated thresholds by round. The interquartile range contracts by six days (86%) between rounds one and three while the median moves two days.

Numbers moved; reasons did not. Between rounds one and two, 24 of 31 panellists changed their nominated figure while 7 changed their primary rationale code. Between rounds two and three the corresponding counts were 19 and 4. Of the 43 recorded figure changes across both transitions, 32 moved toward the median shown to that panellist. Deference-to-panel was the primary code for 2 rationales at round one and 3 at round three, so the movement is not accounted for by panellists reporting that they were deferring.

The contraction did not survive the withheld feedback. At the fourth round the fed arm held an interquartile range of one day and a median of four. The withheld arm returned an interquartile range of three days and a median of five — close to the round-two state, six weeks after the panel had been recorded as agreeing.

Households discard on a different distribution. The diary arm's 1,140 containers are bimodally distributed in every dish category (Figure 2), with an early mode at two days and a late mode past nine. The pooled median discard day is eight. Twenty-one per cent of containers were discarded within a day of the ratified four-day threshold, and a further 21% were discarded on days one or two.

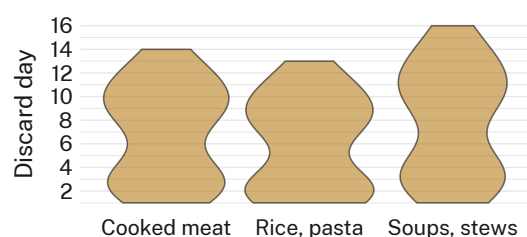


Figure 2: Observed discard day by dish category. Each category is bimodal, with the ratified four-day threshold falling near the waist between the two modes.

Ablation. Recomputing round three with each stratum removed in turn leaves the ratified threshold at four days in every variant and shifts the interquartile range by at most 0.25 days (Figure 3); the differences are not significant. We retain the full panel for completeness.

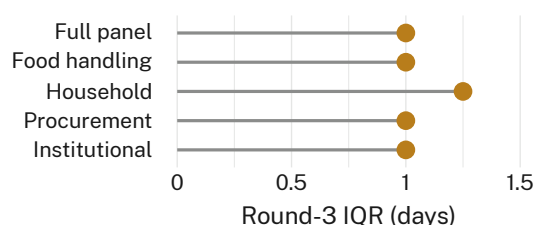


Figure 3: Round-three spread with each stratum removed in turn (rows below the first name the omitted stratum). The consensus criterion is satisfied in all five variants.

Discussion

The procedure did what a Delphi is asked to do: it produced a defensible number under a criterion set in advance, and the number has since been tabled and entered on the Horizon Register as a settled definition. We observe consistent convergence across strata, and the leave-one-out results suggest the threshold is not an artefact of any single constituency.

Two observations qualify the reading. The rationale coding indicates that the panel's agreement is carried by movement in the figures rather than movement in the reasoning behind them, which is consistent with what has been reported of iterated estimation under social influence [15] and with the narrowing this institution has previously observed between a scan entry and its minute [17]. The withheld-feedback round is consistent with the same reading. Meanwhile the diary arm's median sits at twice the ratified figure, which may indicate that households are applying a standard the panel has not yet captured, or that the diary period was unrepresentative. We would not want to over-interpret either possibility on a single instrument.

Limitations

The panel was drawn from four strata and one jurisdiction, though the consistency of the contraction across all four strata suggests the procedure would behave similarly elsewhere. The diary arm ran for 28 days, which constrains the observable upper tail, although the late mode

was already well resolved within the window. The rationale coding rests on what panellists chose to write down; the substantial inter-rater agreement indicates that whatever they wrote was read consistently. We did not test a threshold derived from the diary data against the panel, an asymmetry we regard as appropriate given that the panel was the instrument under study.

Conclusion

A four-stratum expert panel, run over three rounds under a pre-registered consensus criterion, ratified a four-day threshold for household leftover discard. The criterion was met by a six-day contraction in spread accompanied by a two-day movement in the median, with panelists revising their figures substantially more often than their stated reasons, and the contraction did not persist when the group median was withheld. Households in the parallel diary arm discarded at a median of eight days on a bimodal distribution. Future work will extend the instrument to frozen storage, and consider whether the ratified threshold should be re-ratified on a standing cycle.

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